

Antan Roy

+91-8420509687 | Kolkata, India | antanroy2002@gmail.com | linkedin.com/in/antanroy | [Portfolio](#) | github.com/antan2002

Experience

Nrolled – Workforce Management

Dec 2025 – May 2026

Software Engineer Intern

Remote, Canada

- Delivered a production **React Native + TypeScript** staffing platform covering dual customer/worker flows, real-time DB sync, and push notifications; implemented **unit and integration tests** achieving **90%+ code coverage** on critical booking and assignment flows.
- Modeled a **Postgres trigger-driven** enum state machine eliminating all polling; validated **12+ state transitions** with automated test suites, ensuring zero regression across deployments.
- Integrated **Stripe payment flows** into the platform, building end-to-end billing, invoicing, and transaction management pipelines in production.

C & N Green Energy Pvt. Ltd

Aug 2023 – Nov 2025

Embedded Engineer

Kolkata, India

- Revamped EV charger fault-detection in **C++** using interrupt-driven sensor monitoring; stress-tested **10+ fault edge cases** through integration suites, slashing detection latency by **30%**.
- Constructed a **WebSocket** protocol between embedded devices and CMS; tuned ACK batching via **latency profiling**, compressing end-to-end response time from 500ms to **100ms**.

Projects

IntelliCart: AI E-commerce Platform | Python, React.js, Node.js, Gemini-Pro, MongoDB | [GitHub](#)

Jul 2025

- Launched a scalable full-stack e-commerce platform with a **Python** backend and React.js frontend powered by **Gemini-Pro** AI recommendations across **1,000+** catalog items; deployed to production via Render.
- Hardened LLM output correctness with **Zod** schema validation; authored **unit tests** spanning 10+ recommendation edge cases, eliminating free-text drift and keeping every result traceable to a real product.

AI Trip Generator | React.js, Node.js, Gemini AI, Google Maps API | [GitHub](#) | [Live](#)

Dec 2024

- Shipped a scalable trip planner where **Gemini LLM** converts user preferences into structured multi-day itineraries via a **Node.js** API and **React** frontend; segmented service layers for independent testability.
- Wired **Google Maps Places API** for live POI data and **Distance Matrix API** for per-route cost estimation; verified API contracts through **automated functional tests** across 15+ itinerary scenarios.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, SQL, C++, HTML/CSS

Backend: Node.js, Express.js, REST APIs, WebSockets

Databases: PostgreSQL, MongoDB, Supabase

Frontend: React.js, React Native, Tailwind CSS

Tools: Git, GitHub, Postman, VS Code, Vercel

Achievements & Contributions

- Cracked **800+ DSA problems** on [LeetCode](#) and Codeforces; demonstrated analytical depth across graphs, DP, trees, and string algorithms through consistent contest participation.
- Contributed to open-source during **Hacktoberfest 2024** and **GSSoC'24**; merged multiple PRs spanning bug fixes, **test coverage improvements**, and documentation across public repositories.
- langchain-ai/langchain (PR #37653, merged):** Fixed `NotImplementedError` in `Blob.as_bytes_io()` by implementing string-to-bytes encoding for text-initialized blobs, restoring consistent I/O across all langchain-core blob creation paths in a library with **10M+ monthly downloads**.
- elsium-ai (5 PRs merged):** Contributed CI benchmark regression checks, batch deadlock fix on rejected requests, MCP trust framework coverage with **43 tests**, and Zod schema migration from `_def` to `zod-to-json-schema` across the core inference and tooling layer.
- Walmart Sparkathon 2025:** showcased IntelliCart, a scalable AI e-commerce platform with Gemini-Pro recommendations and automated test coverage (React/Node.js/MongoDB).

Education

Maulana Abul Kalam Azad University of Technology (MAKAUT)

Aug 2023 – Jun 2026 (expected)

B.Tech in **Computer Science & Engineering** | CGPA: 7.83/10

Kolkata, India

- Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, object-oriented programming